4 A quantity of gas is trapped by a piston in a cylinder with thin metal walls. The piston is free to move without friction within the cylinder.

Fig. 4.1 shows the cylinder and piston.

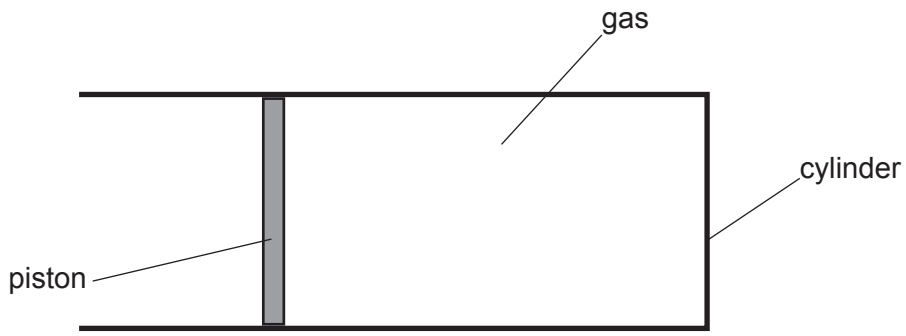


Fig. 4.1

The cylinder is placed inside a freezer.

(a) The air in the freezer is at atmospheric pressure, which is 1.0×10^5 Pa. The area of the piston in contact with the air in the freezer is 2.4×10^{-3} m^2.

(i) Calculate the force exerted on the piston by the air in the freezer.

force = ... [2]

(ii) When the cylinder is first placed into the freezer, the temperature of the gas in the cylinder decreases and the air pushes the piston into the cylinder.

Calculate the work done on the piston by the air in the freezer as the air pushes the piston a distance of 0.021 m into the cylinder.

work done = ... [2]

(b) The initial temperature of the cylinder and the gas is 21 °C and, in the freezer, the temperature of the cylinder decreases to –18 °C.

The thermal capacity of the cylinder is 89 J / °C.

Calculate the change in the internal energy of the cylinder.

change in internal energy = ... [2]

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(c) When the temperature reaches –18 °C, the pressure of the gas in the cylinder is still equal to that of the atmosphere.

Explain, in terms of the particles of the gas, how the pressure remains equal to its original value.

.....

.....

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.....

..... [3]

(d) As the temperature of the metal cylinder decreases, the volume of the metal decreases. The decrease in the volume of the metal is much less than the decrease in the volume of the gas.

Explain, in terms of the particles of the metal, why the decrease in the volume of the metal is less than that of the gas.

.....

.....

..... [2]

[Total: 11]